Arjun Lenan Sandhya

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kavilebhagavathi.github.io/portfolio

Professional Summary

Master's graduate in Computational Engineering at FAU Erlangen-Nürnberg, specializing in High-Performance Computing, parallel and GPU-accelerated programming, and CFD simulation. Industry experience spans structural dynamics data engineering and cloud-based data engineering, bridging numerical simulation with high-performance software development. Proven track record in optimizing critical codebases, delivering scientific analysis pipelines, and delivering fault-resistant software solutions.

Technical Skills

HPC & Parallel Computing: MPI • OpenMP • CUDA (C++ & Python) • ITAC • LIKWID • NVIDIA Nsight & Nsight Compute
Computational Languages: C/C++ •, Python (NumPy, Pandas, SciPy) • SQL • Bash
CFD & Simulation: OpenFOAM • StarCCM+ • OpenLB
3D Modelling: CATIAV5, SolidEdge, Siemens NX, PTC Creo
DevOps & Version Control: Git • GitLab • Azure DevOps • CI/CD Pipelines • JIRA
Cloud Platforms: Microsoft Azure • AWS • Databricks
Languages: Malayalam (Native) • English (C1) • German (A2, actively improving)

Education

Friedrich-Alexander-Universität (FAU), Erlangen-Nürnberg, Germany Oct 2023 – Ongoing
Master's in Computational Engineering

- **GPA:** 2.0 / 4.0
- **Focus Areas:** Computational Fluid Dynamics, High-Performance Computing, GPU programming using CUDA, Parallel Programming using MPI & OpenMP, Lattice-Boltzmann Method
- **Master Thesis:** CFD Simulation of Various Channel Geometries in a River Wave for Rapid Surfing: Ran RANS simulation of an existing river surfing facility and optimized channel geometries for better surf characteristics

A.P.J. Abdul Kalam Technological University, Trivandrum, India Aug 2016 – Sept 2020
Bachelor's in Mechanical Engineering

- **GPA:** 8.8/10.0
- **Bachelor Thesis:** Design and Analysis of an Expandable Extra-Vehicular Activity Module: Designed a novel low-cost EVA module inspired from the principles of origami, focusing on weight-reduction and stowage
- **Focus Areas:** Mechanical Design, Computer-Aided Design (CAD), Fluid Dynamics

Experience

Tasker, Outlier AI Jan 2025 – Aug 2025

- Designed adversarial prompts and ground-truth responses to stress-test advanced AI reasoning models across Python and C++ domains, covering low-level code behavior and physics simulations.
- Engineered failure cases for numerical and logic models, contributing to measurable improvements in model robustness across mathematical reasoning tasks.

Test Vehicle Data Engineering Intern, Rocket Factory Augsburg, Germany Aug 2024 – Oct 2024

- Refactored in-house mechanical analysis software, reducing runtime and memory footprint for the structural engineering team; improved long-term code maintainability.
- Built Python-based scientific analysis pipelines (FFT, signal processing) for sensor data using high-performance libraries (NumPy, SciPy), delivering automated reports for senior management.
- Optimized data processing pipelines for test vehicle telemetry, improving throughput and reliability of engineering data workflows.

Junior Data Scientist, Sentient Scripts Private Limited, Trivandrum, India Jun 2021 – Feb 2023

- Contributed to cloud-based Big Data projects on AWS and Azure, working with scalable and distributed computing architectures. Experience working with both SQL & No-SQL databases.
- Implemented and maintained CI/CD pipelines (Azure DevOps, GitLab), improving deployment reliability in a production environment.
- Coordinated technical team execution and sprint planning via JIRA, supporting on-time delivery of data engineering milestones.

Academic Projects

Parallelization and Optimization of Heat Equation Solver

- Parallelized a finite-difference heat equation solver using OpenMP, achieving significant speedup on multicore CPU architectures.
- Optimized memory access patterns and applied compiler flags with SIMD (AVX) vectorization to maximize cache efficiency and throughput.

HPC Benchmarking of Molecular Dynamics Solver

- Conducted a comparative performance study of an MD solver across CPU (AMD EPYC) and GPU (A10/A100/H100) environments, profiling compute vs. memory-bound bottlenecks.
- Used LIKWID, NVIDIA Nsight, and Nsight Compute to analyze communication overhead, cache utilization, and kernel efficiency.
- Evaluated energy efficiency and throughput metrics to recommend optimal hardware configurations for large-scale physical simulations.

Particle Dynamics Solver (Ongoing)

- Involved in writing a particle dynamics solver in C++ to execute in both CPU and GPU based environments.
- Implemented Lennard-Jones potential and neighbor-list logic. Currently working on periodic-boundary conditions, and energy conservation.

Achievements

Competition on Sustainable Agriculture 2026 (Budapest): Led team to **2nd Place** in CompSA 2026 hackathon; designed a framework and gamified app for flash flood prevention/prediction for the village of Pilisszántó.

SONY x SIEMENS Realize Live 2025 (Amsterdam): Led the FAU university team in the international Student Design Hack competition. Developed a product design for an immersive entertainment system.

SIEMENS x BEST Engineering Competition 2024: Led a four-member team to **3rd Place** in Engineering Design using SolidEdge; project featured in the SIEMENS newsletter and incorporated into the SolidEdge global outreach program.

NASA SpaceApps Challenge 2020: Conceptualized a novel In-Space Docking and Assembly System; submission shortlisted for the Global Finalist selection list.

Best Student Project (S.A.M.E. 2021): Part of the team that developed an award-winning bamboo-based chassis and body for an ultra-mileage challenge vehicle.